

Physical Biology of Transcriptional Control

Neuroscience School of Advanced Studies

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Introduction

The ability to measure the full complement of RNA molecules in space and time within individual cells—and to interpret the torrent of data emerging from these technologies—is transforming how we dissect biological function. In this series of hands-on lectures, we will explore how physical modeling can engage in a dialogue with these new data modalities to uncover the molecular mechanisms governing transcription and its regulation. Participants will be introduced to the basics of dynamical systems theory and chemical master equations to predict both the mean and distribution of mRNA over time. We will also develop a “theory of the experiment” for single-cell sequencing, examining how to extract meaningful mechanistic insights from complex datasets. The sessions will combine pen-and-paper derivations with simple coding exercises in Google Colab, accessible from any laptop or tablet. No prior background in mathematics, physics, or programming is required.

Course Materials

All materials we will use throughout our lectures can be found at the **Course Website**.

Instructors and TAs

Instructor:

- Hernan Garcia (hggarcia@berkeley.edu)

TA:

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Tentative syllabus

- **Lecture 1: Order of Magnitude Biology**
 - Street-Fighting Mathematics: Order-of-magnitude estimates as a tool for discovery in the living world
- **Lecture 2: Stuff(t) or the Protocol of Biological Dynamics**
 - Time evolution in biology.
 - Update rules.
 - Transcription of the constitutive promoter.
- **Lecture 3: Null Hypotheses in Biology: The Poisson Distribution.**
 - The Great Probability Distributions of Biology.
 - The Poisson distribution: Bombs over London, constitutive promoters and sequencing the human genome.
- **Potential Extra Lectures: The Synthetic Transcriptome**
 - The single-cell sequencing revolution.
 - The physics of single-cell sequencing.
 - Dimensionality reduction for the transcriptome.

Online Resources

- Physical Biology of the Cell @ Berkeley Courses.
 - Here, you can find links to previous versions of the Bootcamp as well as the semester-long course on Physical Biology MCB237. These include recorded lectures, class notes, reading materials and Python exercises.
- Physical Biology of the Cell @ Berkeley YouTube Channel

- Compilation of recorded lectures on Physical Biology. New lectures will be added throughout the bootcamp.
- Introduction to Data Analysis in the Biological Sciences @ Caltech
 - This is an excellent and comprehensive course on Python and data analysis in the biological sciences by Justin Bois at Caltech.
- W3 Python Tutorial.
 - While we will not expect any prior Python knowledge, this online tutorial could come handy for those that want to start delving into Python or who need a refresher.

Suggested Reading

- Phillips, R. *et al.* (2012). *Physical Biology of the Cell*, 2nd Edition. Garland Science.
 - Our course will be loosely based on this textbook. Note that we are actively working on the 3rd edition and I might hand out some chapters as we visit their topics in lecture.
- Kinder, J. and Nelson P. (2018). *A Student's Guide to Python for Physical Modeling: Updated Edition*. Princeton University Press.
 - This book is a great introduction to programming in Python. It could be useful reference material throughout the semester.
- Alberts, B. *et al.* (2014). *Molecular Biology of the Cell*. W. W. Norton & Company.
 - This book will be particularly useful for those needing a refresher of biology.
- Milo, R. and Phillips, R. (2015). *Cell Biology by the Numbers*. Garland Science.
 - This book, together with its companion BioNumbers website, has become the reference source for biological numeracy. The book can also be downloaded from book.bionumbers.org.

- Mahajan, S. (2010). Street-Fighting Mathematics: The Art of Educated Guessing and Opportunistic Problem Solving. MIT Press.
- Weinstein, L. and Adam, J.A. (2008). Guesstimation: Solving the World's Problems on the Back of a Cocktail Napkin. Princeton University Press.
 - These two books are fantastic resources for those wanting to learn more about estimation writ large.